

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
M.Tech CSE	Indian Institute of Science, Bangalore	IISc, Bangalore	7.4	2022-2024
B.Tech CSE	Aligarh Muslim University, Aligarh	AMU, Aligarh	9.193	2018-2022

EXPERIENCE

- IBM

Aug 2024 - Present

Software Specialist

Bangalore

– At IBM, I worked on solving complex routing challenges as part of the Routing Development team, collaborating closely with US-based colleagues. Leveraging advanced C programming, data structures, and computational geometry knowledge, I contributed to developing high-performance solutions tailored to meet intricate routing requirements, ensuring both efficiency and scalability.
- LLM QA Enhancement with Knowledge Graphs – British Telecom and SML Lab

Aug 2023 - July 2024

Statistics and Machine Learning Lab Student

IISc Bangalore

– Collaborated with British Telecom to improve LLM question-answering using knowledge graphs.

– Experimented with models like GreasLM and UniKGQA, based on recent research.
- Samsung Research Institute

Feb 2022 - June 2022

Software Engineer Intern

Noida

– Worked as a Software Developer in the Sensor and GPS R&D Team.

RECENT PROJECTS

- Free-Hand Routing

Apr 2025 – Jul 2025

IBM, Bangalore

– Introduced an intuitive **free-hand routing paradigm** that allows designers to sketch routing topologies naturally—like drawing on a whiteboard, notepad, or digital canvas—while the system automatically translates them into **DRC-clean, manufacturable layouts**.

– Seamlessly bridges the gap between **conceptual design intent and automated layout generation**, significantly reducing manual effort and design turnaround time.

– **Patent under filing** for the core methodology and implementation.
- Routing Quality Checker

Jan 2025 – Mar 2025

IBM, Bangalore

– Designed and developed a **Routing Quality Checker** to systematically detect routing quality degradation and violations in chip layouts, and recommend **timing- and noise-aware fix-up strategies**.

– Addressed the challenge of analyzing **millions of multi-layer 3D geometries**, formulating the problem as a computationally hard optimization task and implementing multiple complementary algorithms to collaboratively derive **near-optimal, DRC-clean solutions** at scale.
- Short Fixing in Chip Design (C++)

Aug 2024 – Dec 2024

IBM, Bangalore

– Designed and implemented a high-performance **3D sweep-line algorithm** to accurately detect short-circuit violations across large-scale chip layouts.

– Developed a **timing-aware rerouting framework** to resolve wire shorts without introducing secondary violations, efficiently navigating **millions of 3D geometries** to compute near-optimal, manufacturable routing paths.
- Data Analytics Projects

Jul 2023 – Dec 2023

Data Analytics, IISc Bangalore

– Executed end-to-end data analytics projects spanning multiple domains, including sports analytics (Duckworth–Lewis–Stern method), healthcare (smoking impact, COVID-19 modeling, color blindness), neuroscience (visual processing), space analytics (Mars orbit analysis), social networks (community detection), and machine learning (hypothesis testing, recommender systems, NLP).
- Deep Learning with Natural Language Processing

Jul 2023 – Dec 2023

DLNLP, IISc Bangalore

– *Sentiment Analysis using LSTM*: Built a scalable sentiment classification pipeline using **LSTM networks** trained on large-scale textual review datasets.

– *Unified Date Format via Neural Machine Translation*: Implemented an **encoder–decoder model with attention** to normalize heterogeneous date formats into a unified canonical representation.

- **Optimizing Power Consumption on Smartphones using Machine Learning**

Feb 2022 – Apr 2022

Samsung Research Institute, Noida

- Independently collected and preprocessed large-scale **time-series sensor data** (ambient light, gyroscope, accelerometer) to model real-world user interaction patterns.
- Developed a **Naive Bayes-based usage state classifier** to infer active/inactive device states, enabling adaptive system behavior for improved **power efficiency**.

THESIS

- **M.Tech Thesis: Hallucinations in Large Language Models**

Jan 2024 – Jun 2024

Indian Institute of Science (IISc), Bangalore

- Investigated hallucination phenomena in large language models, with a systematic study of **prompting strategies** and **Retrieval-Augmented Generation (RAG)** techniques using the **MedHALT dataset**.
- Designed controlled experiments to analyze factual inconsistency, quantified hallucination trends, and evaluated mitigation effectiveness across different inference setups.
- Thesis Link: [View Thesis](#)

- **Enhancing Question Answering with Large Language Models**

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IISc Bangalore in collaboration with British Telecom

- Conducted collaborative research with **IISc Bangalore** and **British Telecom** on improving **question answering (QA) performance in LLMs** through the integration of **knowledge graphs** and advanced retrieval strategies.
- Explored and benchmarked multiple **LLM architectures** and **QA datasets**, evaluating techniques such as structured knowledge grounding, retrieval-augmented generation, and hybrid symbolic-neural approaches to improve factual accuracy and reasoning.
- Project Report: [View Report](#)

- **B.Tech Thesis: Movie Recommendation System**

Aug 2021 – Apr 2022

Aligarh Muslim University

- Built an end-to-end movie recommendation system combining **content-based filtering** and **collaborative filtering** techniques on MovieLens data.
- Addressed real-world challenges such as **data sparsity**, **cold-start**, and **search ambiguity**, and deployed an interactive web interface using **Django**.
- Project Report: [View Report](#)

SKILLS

- **Technical Skills:** Machine Learning, NLP, Data Modeling, Data Mining, Algorithms and Data Structures
- **Soft Skills:** Critical Thinking, Problem Solving, Time Management, Leadership
- **Programming Languages:** C/C++, Python
- **Database:** Advanced SQL, Firebase